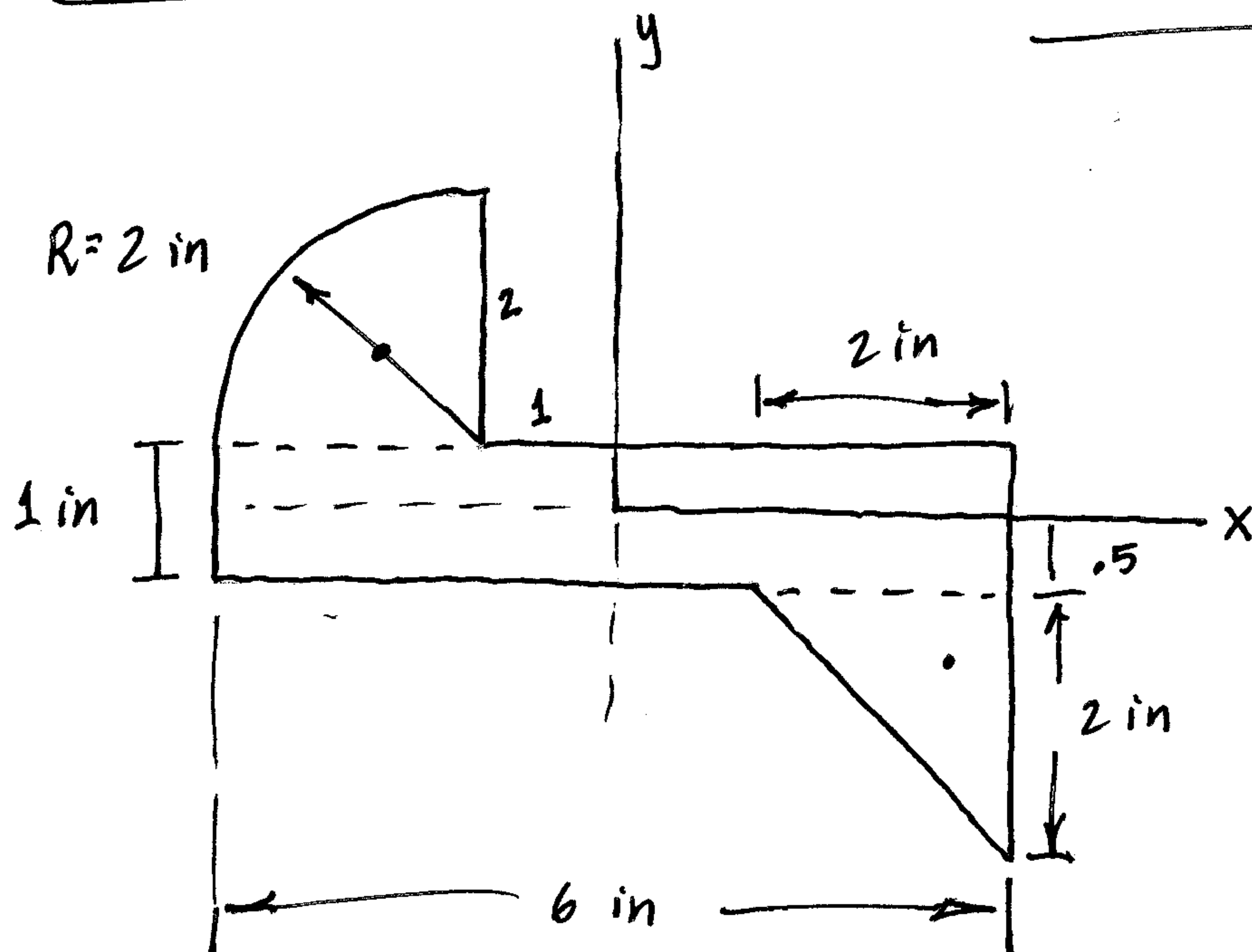


POI Ex Prob 2

Determine I_x, I_y, I_{xy}



$$\frac{4R}{3\pi} = \frac{4(2)}{3\pi} = .8488 \text{ inch}$$

$$I_{\frac{2}{3}} = .667$$

$$I_x = \left[\frac{1}{12}(6)(1)^3 \right]_{\square} + \left[.05488(2)^4 + \frac{\pi(2)^2}{4} (.5 + .8488)^2 \right]_{\Delta}$$

$$+ \left[\frac{1}{36}(2)(2)^3 + \frac{1}{2}(2)(2)(.5 + .667)^2 \right]_{\nabla} =$$

$$= .5 + 6.593 + 3.17$$

$$I_x = 10.26 \text{ in}^4$$

$$I_y = \left[\frac{1}{12}(1)(6)^3 \right]_{\square} + \left[.05488(2)^4 + \frac{\pi(2)^2}{4} (1 + .8488)^2 \right]_{\Delta}$$

$$+ \left[\frac{1}{36}(2)(2)^3 + \frac{1}{2}(2)(2)(3 - .667)^2 \right]_{\nabla}$$

$$= 18 + 11.62 + 11.33$$

$$I_y = 40.95 \text{ in}^4$$

$$I_{xy} = \sum (\bar{I}_{x'y'} + A dx dy)$$

$$= [0 + 0]_{\square} + \left[+.01647(2)^4 + \frac{\pi(2)^2}{4} (-1.8488)(.5 + .8488) \right]$$

Positive! dx $dy = 1.3488$

$$+ \left[-\frac{1}{72}(2)^2(2)^2 + \frac{1}{2}(2)(2)(3 - .667)(-.5 - .667) \right]$$

Neg! $dx = 2.33$ $dy = -1.167$

$$I_{xy} = 0 - 7.57 - 5.67$$

$$I_{xy} = -13.24 \text{ in}^4$$